

Stock Market Price Modeling Using Geometric Brownian Motion: A Study on NEPSE Index Behavior

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Abstract

The stock market is uncertain and influenced by many economic and social factors, making accurate prediction difficult. The main objective of this project is to study stock price behavior using a stochastic model known as Geometric Brownian Motion (GBM).

In this study, historical data of the Nepal Stock Exchange (NEPSE) index is analyzed to estimate the drift and volatility parameters of the GBM model. Using these parameters, stock price paths are simulated and compared with actual market movements.

The results show that Geometric Brownian Motion is able to capture the random and continuous nature of stock price changes. Although it cannot predict exact future prices, the model provides useful insight into market behavior and risk. This study highlights the importance of stochastic models in understanding stock market dynamics.

1 Background

The stock market is a platform, physical or online, where investors buy and sell company shares. Share prices fluctuate based on supply, demand, and market expectations, giving rise to bullish (rising) and bearish (falling) trends. The figure below shows the establishment of stock exchanges, i.e., the regulatory and managing bodies of the stock market in different countries.

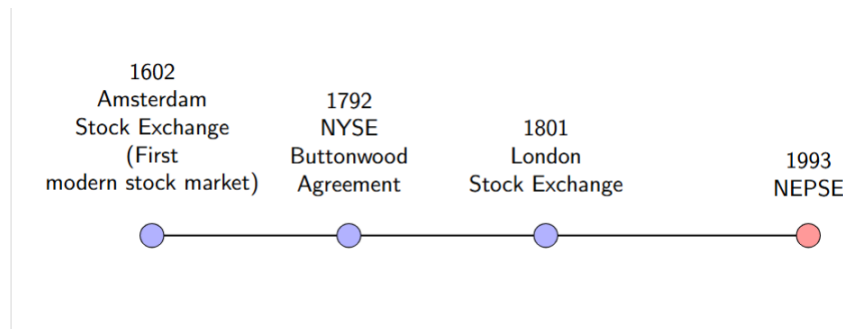


Figure 1: Origin of the Stock Market

1.1 NEPSE Index

The NEPSE Index is the primary indicator of market performance in Nepal, aggregating price movements from sectors such as banking, insurance, hydropower, manufacturing, and trading. It is market capitalization-weighted, with a base value of 100 points in 1994/95. Daily NEPSE closing prices from 2020 to 2025 are converted to logarithmic returns for continuous-time modeling.

Due to political events, policy changes, and macroeconomic factors, the NEPSE exhibits high volatility, making it suitable for stochastic modeling. This study applies the Geometric Brownian Motion (GBM) model to simulate the index and estimate key parameters like drift and volatility.

1.2 Study Objectives

The objectives of this study are to:

1. Model and simulate the NEPSE Index using GBM by estimating drift and volatility from historical data.
2. Apply Monte Carlo simulations to generate multiple index paths and analyze their statistical properties.
3. Evaluate GBM performance by comparing simulated paths with actual NEPSE movements, highlighting strengths and limitations.

2 Geometric Brownian Motion (GBM)

Financial asset prices evolve continuously over time under the influence of various economic, political, and market uncertainties. To model this behavior, the Geometric Brownian Motion (GBM) framework is widely used in financial mathematics because it captures both the systematic growth trend of an asset and the random fluctuations observed in real markets [3].

Let S_t denote the asset price at time t . Under the GBM assumption, the price dynamics are governed by the stochastic differential equation

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

where μ represents the drift parameter (average rate of return), $\sigma > 0$ denotes the volatility, and W_t is a standard Brownian motion that introduces randomness into the system.

Applying Itô's Lemma to this equation yields the explicit solution [3]:

$$S_t = S_0 \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right],$$

where S_0 is the initial asset price. This expression implies that S_t remains strictly positive almost surely and follows a lognormal distribution. Such a property makes GBM particularly suitable for modeling financial asset prices, since negative prices are not economically meaningful and return distributions often exhibit multiplicative growth behavior.

2.1 Discrete-Time Approximation and Simulation

In practice, financial data are observed at discrete time intervals (e.g., daily). Therefore, a discrete-time version of the GBM model is used for simulation and empirical analysis:

$$S_{t+\Delta t} = S_t \exp \left[\left(\mu - \frac{1}{2}\sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z_t \right],$$

where $Z_t \sim \mathcal{N}(0, 1)$ are independent standard normal random variables. This formulation enables Monte Carlo simulation, through which multiple potential future price paths can be generated to analyze expected returns, volatility, and risk characteristics.

2.2 Application to NEPSE Index

In this study, daily NEPSE index data from 2020 to 2025 were used to estimate the drift (μ) and volatility (σ). The drift reflects the long-term trend of the index, while volatility measures daily market fluctuations. Using these estimated parameters, Monte Carlo simulations were performed to generate possible future index paths, helping to understand market uncertainty and potential risk.

3 Methodology

3.1 Data Source

The dataset consists of daily closing values of the NEPSE index from 2020 to 2025. Daily data provides a balance between availability and the ability to capture short-term market fluctuations. The data were obtained from the official NEPSE website and preprocessed using a pandas DataFrame. The dataset contains two variables: Date and Index Value.

Table 1: Sample of NEPSE Index Data

Date	Index Value
01-01-2020	1169.50
02-01-2020	1166.21
05-01-2020	1182.31
06-01-2020	1200.69
07-01-2020	1216.92
08-01-2020	1209.60
09-01-2020	1213.11
$\vdots$	$\vdots$
31-12-2025	2633.76

3.2 Preprocessing

Let S_t denote the closing value of the NEPSE index on trading day t . The daily logarithmic returns are computed as

$$r_t = \ln \left(\frac{S_t}{S_{t-1}} \right),$$

where $\ln(\cdot)$ denotes the natural logarithm. Log-returns are preferred due to their time-additive property and consistency with the lognormal assumption underlying GBM.

Table 2: Sample of NEPSE Index Data with Log>Returns

Date	Index Value	Log-Return (r_t)
02-01-2020	1166.21	-0.002817
05-01-2020	1182.31	0.013711
06-01-2020	1200.69	0.015426
07-01-2020	1216.92	0.013427
08-01-2020	1209.60	-0.006033
$\vdots$	$\vdots$	$\vdots$
31-12-2025	2633.76	0.004140

3.3 Geometric Brownian Motion Model

The NEPSE index is assumed to follow a Geometric Brownian Motion defined by the stochastic differential equation

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

where μ represents the drift parameter, σ denotes volatility, and W_t is a standard Brownian motion.

3.4 Estimation of Model Parameters

Volatility Estimation

Volatility is estimated as the sample standard deviation of daily log-returns:

$$\sigma = \sqrt{\frac{1}{N-1} \sum_{t=1}^N (r_t - \bar{r})^2},$$

where $\bar{r}$ is the sample mean and N is the number of observations. For the dataset, $N = 1348$ and $\bar{r} = 0.000602$, yielding

$$\sigma \approx 0.01495.$$

Drift Estimation

The drift parameter is estimated using

$$\mu = \frac{\bar{r}}{\Delta t} + \frac{1}{2}\sigma^2,$$

where $\Delta t = 1/252$ represents one trading day as a fraction of a year. This results in

$$\mu \approx 0.15188.$$

3.5 Discrete-Time Approximation

The discrete-time GBM solution is given by

$$S_{t+\Delta t} = S_t \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z_t \right],$$

where $Z_t \sim \mathcal{N}(0, 1)$.

3.6 Monte Carlo Simulation

Simulation steps:

1. Start with the initial price S_0 .
2. Generate a random number Z_t for each time step.
3. Update the stock price using the GBM formula.
4. Repeat to simulate a single path.
5. Repeat the process many times to create multiple possible paths.

Based on the estimated parameters, multiple NEPSE index paths were generated using Monte Carlo simulation. Figure 3 illustrates a representative sample of simulated trajectories, each starting from the observed index level at the beginning of 2026.

4 Results and Interpretation

4.1 Descriptive Statistics of NEPSE Returns

Daily log-returns of the NEPSE index from 2020 to 2025 were computed to examine market behavior. Table 3 summarizes the main statistics.

Table 3: Descriptive Statistics of NEPSE Daily Log-Returns (2020–2025)

Statistic	Index Value	Log-Return
Count	1348	1348
Mean	2268.28	0.000602
Std. Dev.	456.68	0.01495
Min	1166.21	-0.06226
Median	2180.46	-0.00021
Max	3198.60	0.05885

The mean log-return indicates a small positive daily growth, while the high standard deviation reflects substantial volatility typical of emerging markets. Extreme values highlight sharp market movements, suggesting deviations from normality.

4.2 Distribution of Log>Returns

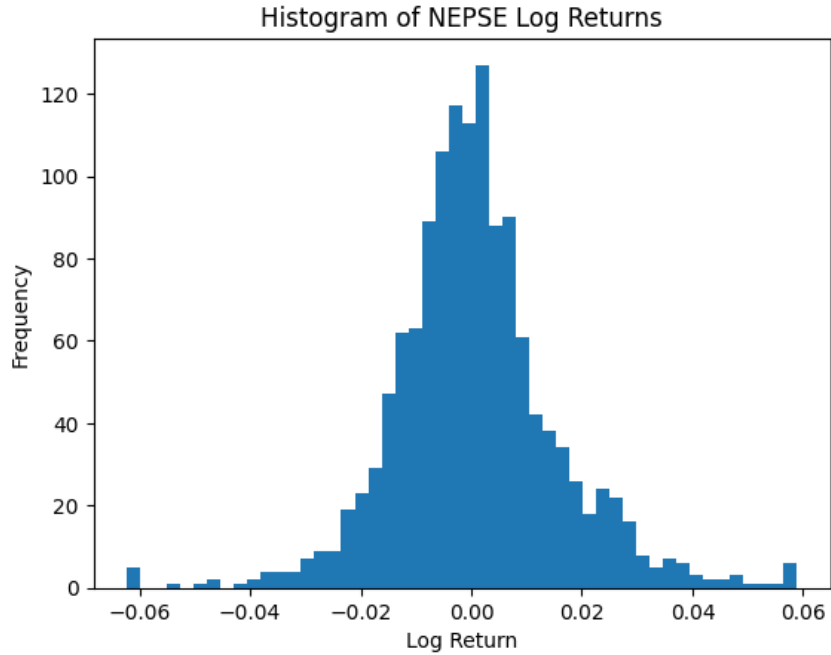


Figure 2: Distribution of NEPSE Log>Returns

The log-return distribution is approximately bell-shaped in the central region, supporting the use of GBM. However, heavy tails indicate that extreme events occur more frequently than predicted by a normal distribution.

4.3 Estimated GBM Parameters

From data, the drift and volatility of NEPSE were estimated (Table 4).

Table 4: Estimated GBM Parameters for NEPSE

Parameter	Estimated Value
Drift (μ)	0.15188
Volatility (σ)	0.01495

The drift indicates a positive long-term trend, while the volatility reflects market uncertainty and sensitivity to economic and policy factors.

4.4 Monte Carlo Simulation of NEPSE Paths

Using the estimated parameters, 1000 NEPSE index paths were simulated via Monte Carlo methods, starting from $S_0 = 2633.76$ (NEPSE closing on 31-12-2025).

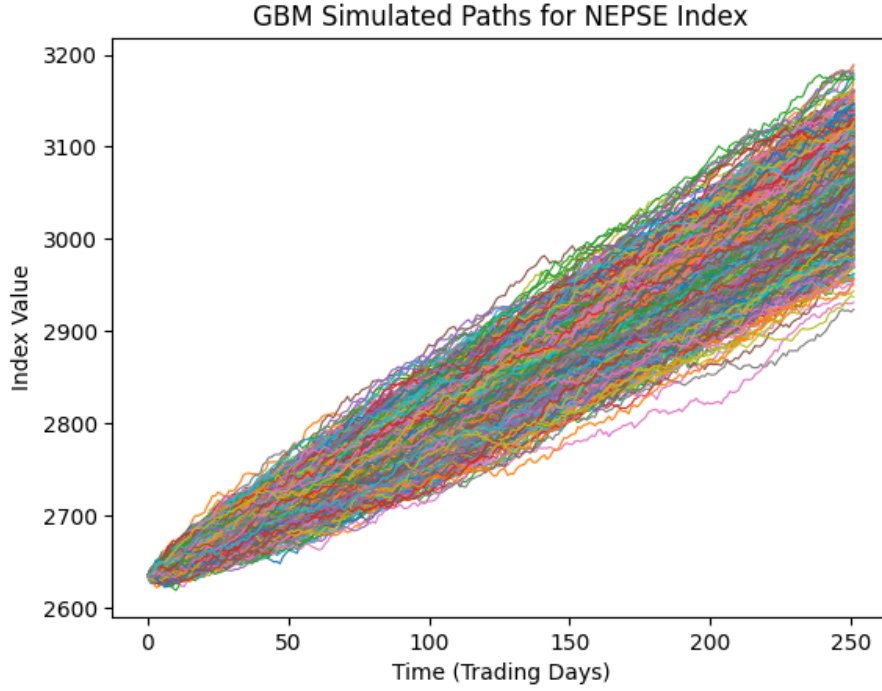


Figure 3: Simulated NEPSE Index Paths Using GBM

The simulation shows all paths remain positive, with dispersion increasing over time, illustrating how forecast uncertainty grows with the time horizon.

4.5 Comparison of Actual and Simulated Returns

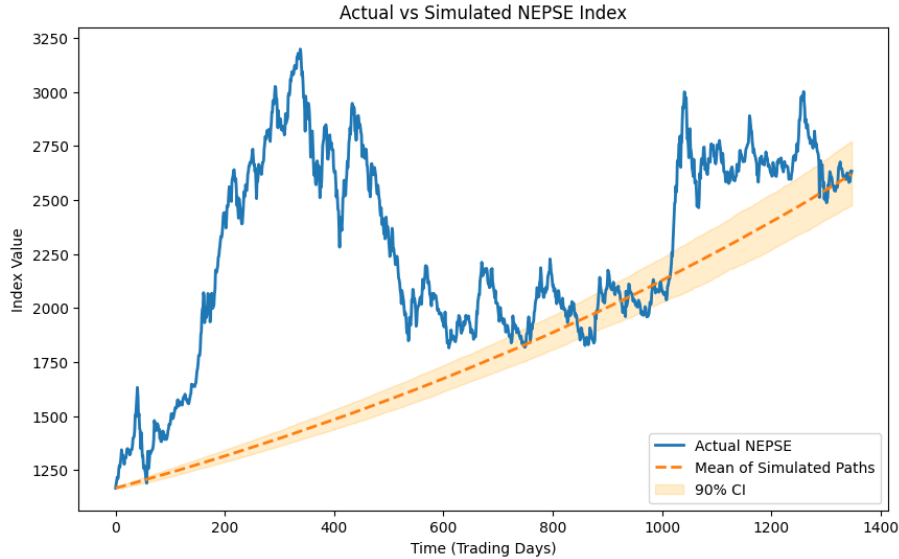


Figure 4: Actual NEPSE Index and Mean Simulated GBM Path

The figure compares the actual NEPSE index with the average path generated from the GBM simulations with estimated parameters and taking initial value i.e., value of 2020/01/01. The blue colour curve represents the historical index, while the dashed line

shows the mean simulated trajectory. The shaded area indicates the 90% confidence interval.

While the simulated path captures the overall upward trend, the actual index shows sharper fluctuations and occasional movements outside the confidence band. This suggests that although GBM reflects the general market direction, it does not fully capture extreme movements and short-term volatility present in real market data.

4.6 Distributional Comparison of Actual and Simulated Returns

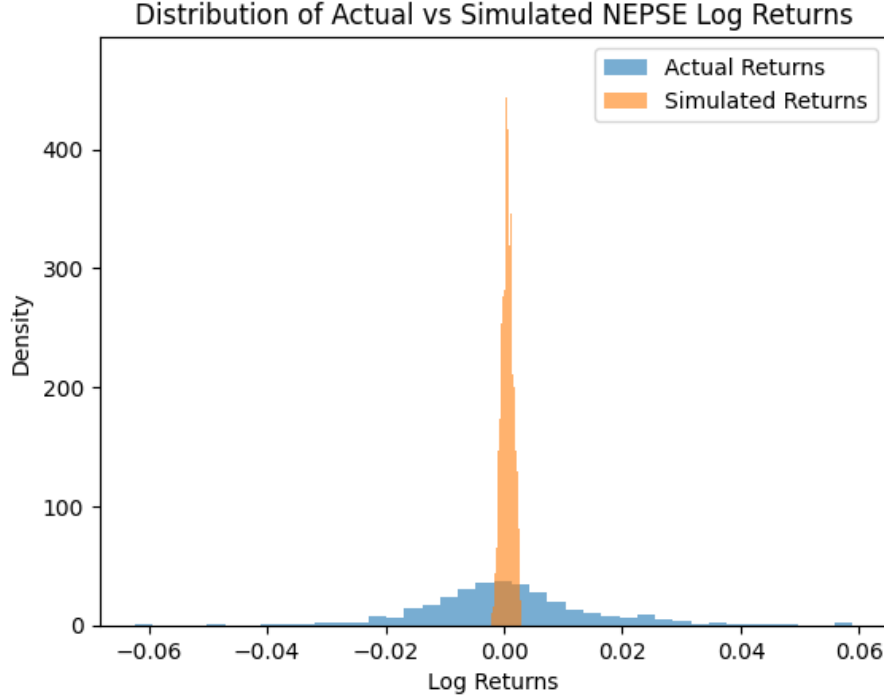


Figure 5: Distributional comparison of actual NEPSE log-returns and GBM-simulated log-returns.

Figure 5 compares the empirical distribution of NEPSE log-returns with those generated by the Geometric Brownian Motion (GBM) model. The observed returns exhibit wider dispersion and heavier tails than the simulated returns, indicating the presence of extreme market movements. While the GBM model captures the central tendency of returns, it underestimates tail risk due to its constant-volatility assumption.

4.7 Model Evaluation

Table 5: First Four Moments of Actual and Simulated Returns

	Actual Returns	Simulated Returns
Mean	0.000602	0.000654
Std Dev	0.014948	0.000947
Skewness	0.238844	0.077803
Kurtosis	2.357758	-0.086356

The GBM model was evaluated by comparing the mean, standard deviation, skewness, and kurtosis of actual NEPSE returns with those from the simulated data. The Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) were found to be 0.0109 and 0.0150, respectively.

These small error values indicate that the model captures the basic statistical properties of the returns reasonably well. However, some differences in skewness and kurtosis suggest that GBM does not fully reflect extreme market movements.

4.8 Expected NEPSE Index Value

The expected NEPSE index after one year under GBM is:

$$\mathbb{E}[S_1] = S_0 e^{\mu} = 2633.76 \times e^{0.15188} \approx 3065.75.$$

This represents a trend-based forecast; actual market outcomes may differ significantly due to volatility and random shocks.

4.9 Limitations of GBM

The GBM model does not capture:

- Volatility clustering,
- Sudden market jumps from economic or political events,
- Time-varying drift and volatility.

These limitations are especially relevant for emerging markets like Nepal.

4.10 Summary

GBM provides a reasonable approximation of NEPSE's average behavior and long-term trend. Monte Carlo simulations illustrate the range of potential index paths and highlight forecast uncertainty, but the model underestimates short-term volatility and extreme events.

5 Conclusion and Recommendations

5.1 Conclusion

This study analyzed the NEPSE index using the Geometric Brownian Motion (GBM) model. The results show that NEPSE has a small positive average return with relatively high volatility, which is common in emerging markets. The GBM model successfully captures the overall upward trend and general stochastic behavior of the index. Monte Carlo simulations further highlight the uncertainty in future price movements.

However, the model does not fully capture short-term fluctuations and extreme market events, mainly due to its assumption of constant drift and volatility. Overall, GBM serves as a useful baseline model for understanding NEPSE dynamics and provides a foundation for more advanced modeling approaches.

5.2 Recommendations

Based on the findings, future research may consider more flexible models such as jump-diffusion or stochastic volatility models to better capture extreme movements and changing market risk. Including macroeconomic factors could also improve predictive performance. While GBM is suitable as a benchmark, it should be complemented with models that account for volatility clustering and sudden shocks. Regular updating of model parameters is also recommended to reflect evolving market conditions.

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